

Image Bundling Revisited: Codec-Specific Savings and a Measured Atlas Selector for Small Web Images

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ABSTRACT

Web pages load many small images; the median mobile page carries 13, and product grids, icon sets, and avatar walls carry far more. Serving each one separately pays per-request overhead plus two byte costs that HTTP/2 multiplexing does not remove: a fixed per-file structural cost (roughly 600 bytes for a JPEG) and the loss of cross-image redundancy. Measured at matched per-tile quality across the three codecs that carry most web images (JPEG, PNG, WebP), bundling many images into one atlas recovers up to 30% of bytes under JPEG and up to 19% under WebP, and the sign of the effect is codec-specific: WebP turns negative on large photographic tiles while JPEG and AVIF never do.

The saving is set by the ratio of per-file overhead to content size, so it falls as tiles grow, from about 30% for 56-pixel photographs to a few percent at 224 pixels. The reversal under WebP is driven by its whole-atlas quantizer rather than by atlasing itself, and it holds across four independent photo populations. Lossless formats lose under a naive grid but win 40–97% under codec-aware packing (vertical strips, shared palettes), so the question is how to pack, not whether. We distill the measurements into a coupling account (savings come from sharing fixed costs; losses from sharing adaptive state) and into an open-source tool that turns a directory of images into deployable bundles and matches an offline oracle on five unseen collections. Cheap source-image features (color, edge, frequency) do not predict the byte effect, but a probe encode of ten to twenty tiles forecasts a group's saving to within two percentage points, so the tool measures candidate bundles rather than modeling them. In a live renderer the pixel atlas is also the fastest to full visibility and the lowest in memory, and a supporting browser study shows the byte savings cut load time on HTTP/1.1 and HTTP/3.

1 Introduction

Product grids, icon sets, avatars, and decorative elements make small images the most numerous resource class on commercial pages. Bundling them into one image, the CSS sprite technique, was standard practice in the HTTP/1.1 era and fell out of favor when HTTP/2 multiplexing removed the per-connection request bottleneck. That reasoning addressed only the request-count cost. Two byte-level costs survive multiplexing untouched: every image file carries a fixed container overhead (headers, quantization tables, ISOBMFF boxes), and every file boundary prevents the codec from sharing entropy context, palettes, and predictors across images.

JPEG, PNG, and WebP anchor the study: together they carry over 70% of images served on the web (HTTP Archive 2024: JPEG 32.4%, PNG 28.4%, WebP 12.0%), decode in every browser, and are where the practical savings live. AVIF, the leading next-generation codec, joins the photographic crossover (Section 5.1) to show the result's codec-dependence against a modern container. The three

formats price the per-file costs very differently: a JPEG file carries, with the libjpeg-turbo encoder and settings used here, roughly 600 bytes of Huffman and quantization table definitions plus headers (the exact figure is encoder- and settings-specific), a PNG a 67-byte structural floor plus per-scanline filter adaptation, and a WebP as little as 30 bytes of container around a heavily image-adapted VP8 payload. This report quantifies what bundling recovers, per codec, tile size, and image count, under a matched-quality protocol, and describes a testbed that measures the timing consequences under HTTP/1.1, HTTP/2, and HTTP/3.

The regimes this study measures map onto concrete page types. Product grids and image-search results serve 100–280-pixel photographic thumbnails by the dozens; recommendation strips, cart previews, and avatar rows serve 48–96-pixel photos; emoji and reaction pickers serve 24–64-pixel flat art by the hundreds, and are one place where sprite atlases remain in production use today (chat applications ship emoji sheets; video platforms ship hover-preview storyboards as frame mosaics). The tile-size axis of the experiments spans exactly this range, so each regime can read its expected saving off the measured curves.

The display side needs no special machinery: a bundled tile is shown by any of CSS `background-position` (universal), `object-view-box` (Chromium), canvas `drawImage` region blits, or SVG `viewBox` cropping. The browser decodes the atlas once and paints windows into it.

Contributions. This report contributes (i) a matched-quality sweep of image atlasing across the three universally supported web codecs (JPEG, PNG, WebP) over tile size and image count, establishing that the byte saving is governed by the ratio of per-file structural cost to content bytes; this is, to our knowledge, the first measurement of atlasing byte savings resolved by codec and tile size at matched perceptual quality (the one prior peer-reviewed sprite study [33] optimized PNG packing geometry with the codec fixed); (ii) a supporting network study of 1,056 validated cold browser loads that separates the timing effect of bundling across HTTP/1.1, HTTP/2, and HTTP/3 under emulated network conditions; (iii) a set of construction techniques with measured effect, PNG vertical-strip packing, explicit and LZ-window duplicate exploitation, chunking for bounded cache invalidation, delta updates via compression dictionaries, and encoder-parameter tuning that flips WebP's photographic-atlas penalty to a gain; (iv) a coupling-spectrum account that unifies the results, savings arise from sharing fixed costs and losses from sharing adaptive state, together with the finding that this adaptive-state penalty is codec-mechanistic and is not predicted by six cheap source-image features, while a small probe encode forecasts it accurately; and (v) an open-source construction heuristic, built on that probe-not-predict principle, that emits deployable bundles from a directory of images.

2 Related work

2.1 Resource bundling and image spriting

Combining many small resources into one is long-standing web practice, catalogued in the performance-engineering literature (Souders [35]) as spriting, concatenation, and DataURI inlining, all of which trade requests for cache granularity in the HTTP/1.1 era; CSS sprites [10] are the image-specific form. The regime they address is real and growing: measurement of page composition shows the modern page is dominated by many small, heterogeneous resources (Butkiewicz et al. [34]), and image requests are the largest class (HTTP Archive [8]). Whether bundling still pays under HTTP/2 multiplexing has been examined mainly for JavaScript and CSS: Khan Academy found unbundled JS slower than bundles under HTTP/2, attributing the gap to worse per-file compression [1], and practi-

tioner analyses reach the same conclusion for concatenated assets [2, 11]. Modern build tools encode the trade-off directly as an inlining size threshold (webpack asset modules, Vite's asset limit). The closest academic work on the packaging question is Marx et al. [29], who test concatenation, embedding, and sharding under HTTP/2 and find the HTTP/1 packaging habits still often help; broader page-load studies show load time is governed by resource dependencies and compute, not bytes alone (WProf [26], How Speedy is SPDY? [27], Polaris [28]), so request-count reductions are only part of the story. The one peer-reviewed study of image spriting itself is Marszalkowski et al. [33], who formulate CSS-sprite construction as a geometric packing problem, model load time, and measure how a PNG sprite's aspect ratio affects its file size, holding the codec fixed at PNG and optimizing layout area. Their axis is packing geometry; ours is codec and tile size at matched quality, which they do not vary. A CSS-Tricks case study reports a 223-icon sprite at roughly 10 KB versus 115 KB unbundled [2], but without protocol-level timing. We find no published measurement that quantifies image atlasing under HTTP/2 or HTTP/3, nor a codec-by-tile-size atlasing sweep at matched quality.

2.2 Protocol performance: HTTP/2 and HTTP/3

HTTP/2 multiplexes many requests over one connection [12] but suffers transport-level head-of-line blocking under loss; HTTP/3 over QUIC [13] removes it with per-stream recovery. Whether the newer protocol is faster in practice is contested and configuration-sensitive: de Saxcé et al. [36] found HTTP/2 not uniformly faster than HTTP/1.1, and rigorous QUIC evaluation shows performance swings widely with implementation and workload (Kakhki et al. [37]). Measurement studies characterize HTTP/2 adoption and page-load behavior [3] and server push [4], and both controlled benchmarks [5] and adoption studies [14] report HTTP/3 gaining most under packet loss while sitting near parity at zero loss. Domain sharding, the historical technique of spreading resources across hosts to widen HTTP/1.1 concurrency [15], is the opposite of bundling; recent measurement finds such HTTP/1-era habits persist even under HTTP/2 [30], which motivates our condition set. This body of work establishes that protocol-level outcomes for a given payload are as much a property of the deployment as of the standard, the context in which our own HTTP/3 concurrency observation (Appendix A.2) should be read; none of it isolates a many-small-images payload against a bundled baseline.

2.3 Small-image coding and multi-image containers

The fixed per-file cost each codec carries is documented through minimal one-pixel files: JPEG XL 24 B, WebP 30 B, PNG 67 B, JPEG 155 B, AVIF 303 B [6, 7]. This overhead makes small images the worst case for heavyweight containers, and production pipelines act on it: Cloudinary declines AVIF for images below 5,000 pixels because the box overhead outweighs the coding gain [16]. Several formats already provide intermediate ways to share structure across many images without a full pixel atlas: JPEG's abbreviated format carries one table-specification datastream ahead of table-less scans [17]; animated WebP stores independently-coded frames in one container [19]; APNG and MNG [40] and the HEIF image-collection format [41] package multiple images in one resource. These occupy the low-coupling end of the spectrum this report maps, sharing container and tables but not the coding model. The JPEG [17], PNG [18], and WebP [19] format definitions specify the table, chunk, and container structures our measurements amortize; JPEG XL adds a modern architecture aimed partly at small images [38]. Matched-quality byte comparison across these formats is established for single images (Google's WebP study measures WebP against JPEG and PNG at equal SSIM [39]) and ex-

tended by peer-reviewed rate-distortion studies against newer codecs [31]; we carry the same matched-quality discipline to the atlas-versus-individual question. Perceptual image-quality assessment, on which our matched-quality protocol rests, is grounded in the structural similarity index [25]. We are not aware of prior work that measures how bundling recovers per-file overhead as a function of codec and tile size.

2.4 Texture atlases and sprite packing

Packing many images into one is standard in real-time graphics, where texture atlases [9, 20, 32] and skyline/MaxRects bin-packing [21] pursue a different objective: production atlas tools (TexturePacker, engine sprite packers) and virtual-texturing systems minimize GPU state changes and packed area, and tune padding for mip-sampling and filtering correctness at tile boundaries rather than for encoded size. The optimization target is binding count and texture memory, not transmitted bytes, so this literature does not subsume a delivery-oriented study: our objective is encoded size and adaptive codec state under matched perceptual quality, which area-minimizing packers neither measure nor optimize.

2.5 Cache granularity and delta delivery

Bundling trades cache granularity for fewer requests, a tension delivery research has long addressed. HTTP delta encoding transmits only the difference between a cached and a current resource (RFC 3229 [42]) using generic differencing formats (VCDIFF, RFC 3284 [43]); shared-dictionary schemes reuse previously delivered bytes as a dictionary for later responses (SDCH [44]), the idea that shared-window and trained-dictionary compressors (Brotli [22], zstd [23]) generalize and that Compression Dictionary Transport [24] revives for the modern web. Standard HTTP caching semantics (RFC 9111 [45]) fix the granularity at the resource, which is exactly what bundling coarsens; we apply delta and dictionary delivery to the whole-bundle cache-invalidation problem (Appendix A.4). HTTP Archive's Web Almanac [8] supplies the population statistics (median 13 images per mobile page, format share) that make the many-small-images regime the common case rather than an edge case.

3 The atlas serving model

3.1 Rendering a tile from an atlas in HTML

A bundled page references one image resource and displays each tile by cropping a window into it. Four standard mechanisms cover every deployment context. The classic and universal one positions the atlas behind a fixed-size element as a background:

```
<div class="tile" style="background-image:url(atlas.webp);  
    background-position:-144px -216px"></div>  
/* .tile has width:72px; height:72px */
```

The element shows the 72x72 region whose top-left corner sits at (144, 216); `background-size` rescales the whole atlas when display size differs from stored size. Chromium additionally supports cropping a real `<img>` element, which preserves alt text, native lazy loading, and `fetchpriority`:

```

```

Where a real image element is needed cross-browser, a fixed-size wrapper with `overflow:hidden` around a negatively-offset `<img>` gives the same window, and SVG gives it declaratively (`<svg viewBox="144 216 72 72"><image href="atlas.webp"/>`). Canvas completes the set for programmatic UIs: `ctx.drawImage(atlas, 144, 216, 72, 72, dx, dy, 72, 72)` blits any region after a single decode. In every mechanism the browser fetches and decodes the atlas once, holds one decoded copy, and paints windows into it; per-tile cost is a paint, not a decode. The experiments use `background-position`, the mechanism with universal support; the four mechanisms differ in image semantics, accessibility, loading control, and browser support, compared for deployment in Table 10 (Appendix A.5).

3.2 Delivery, caching, and memory

An atlas changes the unit of caching from the tile to the bundle. With standard immutable content-addressed URLs (`atlas.3fe2a1.webp`, `Cache-Control: immutable`), an unchanged atlas costs zero requests on a warm cache, and the decode-once property still applies. The cost appears on content change: editing one tile invalidates the whole bundle, so the expected re-download per deploy grows with bundle size. Splitting the collection into k chunked atlases bounds the worst-case invalidation at $1/k$ of the collection while retaining nearly all of the byte and request savings, since chunking adds only a few container headers and a little grid slack, which makes chunking the practical default for collections that update piecemeal. Grouping tiles by update cadence (stable icon set in one chunk, weekly seasonal art in another) further confines invalidation to the chunk that actually changed.

The second resource to budget is decoded memory: a decoded atlas occupies width $\times$ height $\times$ 4 bytes regardless of its encoded size, so a 4096x4096 atlas holds 64 MB of RGBA for a 300 KB transfer. Individual images decode lazily and can be evicted per tile; an atlas is decoded and resident as a unit while any tile is visible. Chunking bounds this cost the same way it bounds invalidation, and below-the-fold chunks combine with lazy loading so off-screen tiles cost neither bytes nor memory.

A live-renderer measurement (Appendix A.3) confirms that at typical tile counts the atlas is in fact the memory-favorable representation, so this analytical caution binds only for very large atlases.

4 Methodology

4.1 Assets and packing

Two deterministic asset classes anchor the study: 521 Twemoji 72x72 flat-art tiles (alpha composited over white) and 521 photographic 224x224 thumbnails, with the photo set additionally downscaled to 112 and 56 pixels for the size sweep and synthetic icon, product-thumbnail, and avatar corpora added for the use-case tests (Section 5.3). Tiles are packed row-major into a near-square grid; a padding variant edge-replicates each tile by 8 or 16 pixels, and a vertical-strip variant packs one tile per row band. All conditions consume identical source pixels.

4.2 Codecs and matched-quality protocol

Following the study's scope, the three dominant web formats encode each condition: libjpeg-turbo JPEG, WebP (lossy and lossless), and PNG, with the lossy codecs swept over a quality ladder $q \in \{30, 50, 65, 80, 90\}$. Quality is the mean over tiles of the per-tile luma SSIM [25], each tile scored after

cropping it back out of the decoded artifact so atlas border bleed is charged to the atlas; the matched target of 0.97 is therefore a mean-tile floor, and Section 5.1 reports the per-tile spread where it is load-bearing. Bytes are compared at equal quality by log-linear interpolation of each condition's rate-distortion curve at the target; the five-point ladder is coarse, so where the atlas and individual curves nearly coincide the interpolation is unstable and equal-quality byte comparison is used instead (Section 5.3). When an atlas's entire ladder exceeds the target, its cheapest measured point is used, understating the atlas's advantage, and the value is reported as a lower bound. Three invariants validate the harness: lossless conditions score SSIM exactly 1.0; an atlas of one image is byte-identical to the individual file; padding never reduces atlas bytes.

4.3 Network testbed

Byte savings are protocol-independent; timing effects are not. The network testbed serves every condition from a Caddy server inside WSL2 with three protocol endpoints (HTTP/1.1, HTTP/2, HTTP/3 on QUIC), shapes real packets with `tc netem` (delay, bandwidth, random loss applied on the server egress), and drives a fresh cold-cache Chromium instance per page load via Playwright. Each page stamps a timestamp after every tile is decoded and two animation frames have painted, giving a single time-to-all-tiles-visible endpoint, and records per-resource transfer sizes and the negotiated protocol from the Resource Timing API; every load is validated against the intended protocol and against the manifest's byte totals, so a page that fetched the wrong way cannot enter the dataset. Four serving conditions run: N individual files, one atlas, four chunked atlases, and a byte-bundle (the N encoded files concatenated into one binary resource plus an offset index; the client slices the buffer and decodes each tile from its own bytes, retaining per-file codec adaptation while collapsing N requests into one). The reported timing study fixes these at N = 500 tiles (the most demanding count), both asset classes as WebP q80, three protocols, and four network profiles, giving 96 cells. Within each profile the load order is randomized across conditions, protocols, and repetitions, and the first repetition of each cell is discarded as a warm-up, leaving 11 measured loads per cell (1,056 in all).

4.4 Artifact availability

All code, asset manifests, raw per-run measurements, and the construction heuristic are released at github.com/ApartsinProjects/ImageBundling, and every table and figure in this report is regenerated from that data by a single build command. The measurements use Pillow 12.2 (libwebp 1.6.0, libjpeg-turbo via libjpeg 8.0, zlib-ng 1.3.1), Python 3.14, zstandard and Brotli for the delta-update study, Caddy 2.11.4 for the three-protocol server, and Playwright 1.58 driving Chromium 145 for the network loads. The photographic corpus is drawn from Lorem Picsum and the flat-art corpus from the Twemoji set; both frozen manifests are in the repository so every condition consumes identical source pixels.

5 Results

5.1 Byte savings at matched quality



Figure 1. A 100-tile atlas (10x10 grid of 72x72px tiles, 720x720px). Encoded with identical encoder settings both ways (distinct from Table 1’s matched-quality protocol), the 100 tiles cost 147,361 bytes as separate JPEG files and 112,059 bytes as this single JPEG (24% less), while 100 HTTP requests become one. Each tile is displayed with CSS background-position.

Table 1 reports the saving from atlasing at SSIM 0.97 across both classes. Three regularities organize the table. First, savings scale with the ratio of per-file structural cost to content bytes: 72-pixel flat-art tiles encode to 1–3 KB, so JPEG’s roughly 600 bytes of per-file tables and headers alone account for most of its measured 19–26% saving across N, while WebP’s 30-byte floor leaves it the smallest lossy gain (8–15%). Second, savings broadly increase with N and flatten by a few hundred tiles as amortization completes (the trend is not strictly monotonic; because each N draws one deterministic subset, small non-monotonicities such as WebP flat art at 8.3/15.2/8.2% for N = 50/200/500 reflect subset composition, not a scaling law). Third, tile size, not image count, is the dominant variable, and the photo rows of Table 1 trace the full crossover: downscaling the same 500 photographs from 224 to 112 to 56 pixels moves the JPEG saving from 3.0% to 9.8% to 29.8%, and moves WebP from –8.5% through –1.4% to +15.3%, placing WebP’s bundling break-even near 100-pixel tiles. Search-result and

recommendation thumbnails (48–120 px) therefore sit squarely in the paying regime for both formats, while product-grid images (200 px and up) pay only under JPEG, and only a few percent.

To confirm the crossover is not an artifact of the single deterministic subset each Table 1 cell uses, we resampled it: for each (tile size, N, codec) we drew 20 random N-tile subsets from the full pool and computed the matched-quality saving of each. The medians track Table 1 and the 95% bootstrap intervals are tight and separate the regimes. For photos the JPEG saving is 29.1% (95% CI [28.4, 29.5]) at 56 px, 9.3% [8.9, 9.8] at 112 px, and 2.6% [2.4, 2.8] at 224 px, all clearly positive; WebP is 14.8% [10.0, 17.4] at 56 px, straddles zero at 112 px ([−1.6, 1.9]), and is clearly negative at 224 px (−8.0% [−12.0, −6.7]). The WebP break-even near 100 px is thus the point where its interval crosses zero, not a single-sample coincidence.

Table 1. Bytes saved by atlasing (%) at matched per-tile SSIM 0.97 (lossy codecs; interpolated on the quality ladder) or exact lossless comparison (-ll columns), grid packing, no padding. Negative values (red) mean the atlas is larger; n/a marks a cell whose quality ladder lies entirely above the target with no interpolable point.

class	N	JPEG	WebP	PNG	WebP-lossless
flat art 72px	10	19.4	n/a	4.3	-0.2
flat art 72px	50	26.3	8.3	5.4	5.4
flat art 72px	200	25.1	15.2	-3.1	-4.7
flat art 72px	500	26.3	8.2	-5.7	-3.6
photos 56px	10	22.3	15.3	-1.4	2.9
photos 56px	50	27.7	14.2	1.1	2.5
photos 56px	200	29.3	18.8	0.4	3.7
photos 56px	500	29.8	15.3	0.8	4.9
photos 112px	10	6.3	2.7	-3.8	1.8
photos 112px	50	8.4	1.7	-0.9	4.3
photos 112px	200	9.4	-0.1	-1.8	4.3
photos 112px	500	9.8	-1.4	-1.6	1.4
photos 224px	10	1.0	-4.3	-5.7	-3.4
photos 224px	50	1.8	-5.1	-2.8	3.0
photos 224px	200	2.6	-8.2	-3.8	0.3
photos 224px	500	3.0	-8.5	-3.6	-4.9

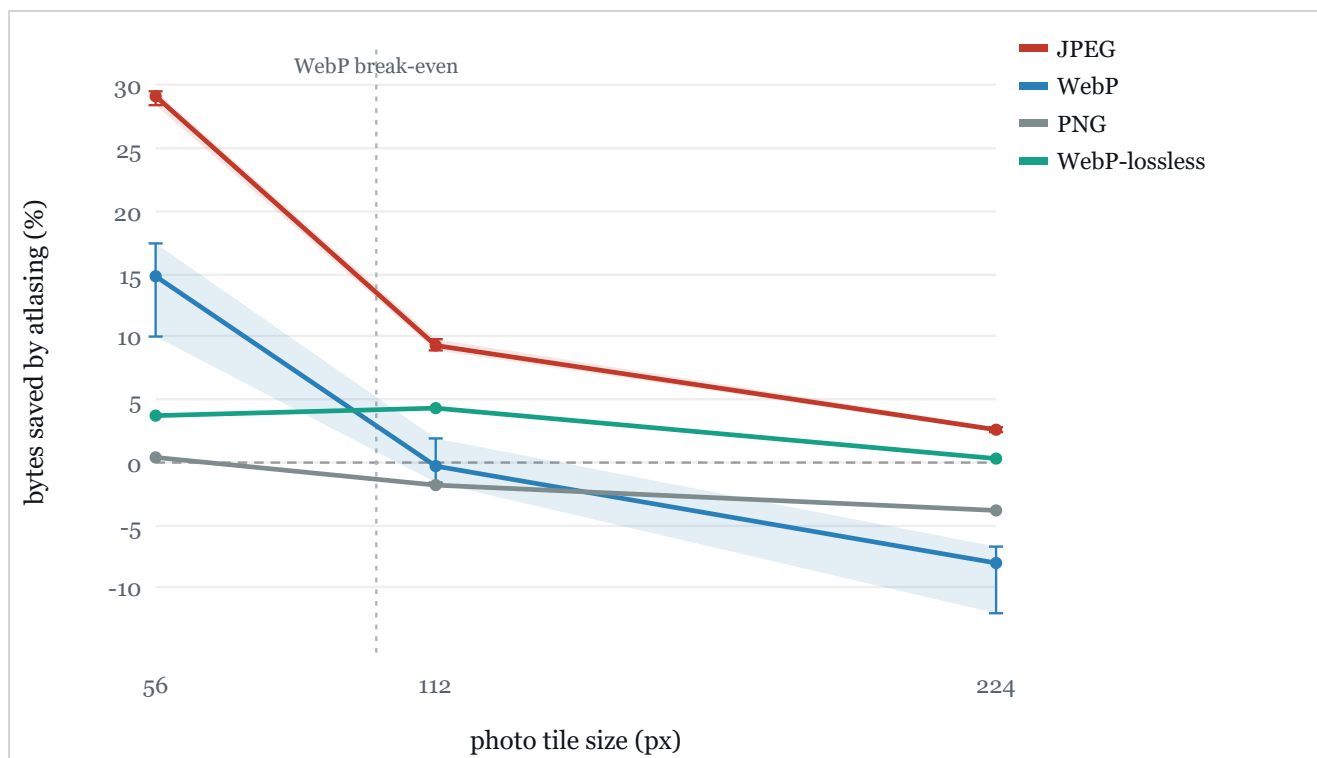


Figure 2. Byte savings from atlasing 200 photographic tiles at matched quality, as tile size grows from 56 to 224 pixels. JPEG and WebP are plotted as medians of 20 random subsets with 95% bootstrap confidence intervals (whiskers and shaded band); the lossless formats are exact. The saving tracks the per-file-overhead-to-content ratio and so falls with size: JPEG stays clearly positive throughout, while WebP's interval crosses zero near 100 px (dashed marker) and is clearly negative by 224 px. Table 1 gives the full grid over image count and the flat-art class.

Atlasing is not free where per-image adaptation matters. On flat art, both lossless formats lose from atlasing at $N \geq 200$ (PNG -3 to -6% , lossless WebP -4 to -5%): PNG chooses its prediction filter per scanline and an atlas scanline crosses dozens of unrelated tiles, while lossless WebP fits transforms and entropy codes per image, and one global model over hundreds of heterogeneous tiles cannot match hundreds of specialized ones. Lossy WebP shows the same inversion on photographic tiles: VP8 adapts entropy tables per image and allows at most four quantizer segments per frame, so an atlas of 500 diverse photos shares four segments where individual files had four each. This penalty is a quality effect, not a byte cost, and the -8.5% figure decomposes into the pair it comes from: at equal encoder quality the 500-photo WebP atlas costs essentially the same bytes as individual files (-0.2% at q80) but reaches a slightly lower mean per-tile SSIM (0.9748 vs 0.9784, a 0.004 deficit); the matched-quality protocol prices that small deficit as the -8.5% through interpolation on a steep rate-distortion curve. The sign is robust (the deduplicated-subset bootstrap interval is $[-12.0, -6.7]$), but the magnitude is metric-dependent, and Appendix A.1 shows two encoder flags reverse it to $+5\%$. Edge-replicated padding costs roughly 7 percentage points of saving per 8-pixel step for JPEG and roughly 10 for WebP (flat art, $N = 200$), pricing the block-alignment mitigations against chroma bleed.

The absolute comparison compounds codec choice with bundling: at $N = 500$ and SSIM 0.97 on flat art, individual JPEG files cost 630 KB, the JPEG atlas 464 KB, individual WebP files 240 KB, and the WebP atlas 220 KB. Moving a legacy individual-JPEG deployment to a WebP atlas cuts bytes by 65%; the format change contributes most of it, and bundling contributes the rest while also collapsing 500 requests into one.

Matched-quality comparison equalizes the mean, not the tails. Table 2 shows the per-tile SSIM distribution: the WebP atlas systematically carries a worse worst tile than individual files (a flat-art tile stuck at 0.85 that no quality setting recovers), while JPEG's atlas and individual tails coincide. A practitioner enforcing a hard per-tile floor rather than a mean should treat the WebP pixel atlas accordingly, or use the byte-bundle, which preserves each tile's own encoding.

Table 2. Per-tile SSIM distribution (mean, 5th percentile, minimum) for individual files versus the grid atlas at equal encoder quality (q80, N=500). Matching the mean can still leave the atlas with a worse worst tile: the flat-art WebP atlas has a tile stuck near 0.85 that no quality setting recovers, a shared-segment boundary casualty, whereas JPEG's atlas and individual tails coincide.

condition	ind mean	ind p5	ind min	atlas mean	atlas p5	atlas min
flat art 72px, WebP	0.991	0.984	0.954	0.984	0.960	0.853
flat art 72px, JPEG	0.979	0.967	0.946	0.979	0.967	0.957
photos 56px, WebP	0.979	0.961	0.925	0.980	0.963	0.918
photos 224px, WebP	0.978	0.965	0.942	0.975	0.960	0.926

The crossover is not an artifact of the luma-SSIM metric. We re-measured the photo savings with SSIMULACRA2, the reference libjxl perceptual metric, scoring each condition on the full grid image at matched quality (Table 3). JPEG's atlas saving stays positive at every tile size (about +33% at 56 px, +12% at 112 px, and +3% at 224 px at a matched score of 70), tracking the luma-SSIM result closely. WebP is positive only for the smallest tiles (+12% at 56 px) and turns clearly negative by 112 px (-24%) and 224 px (-29%). The break-even therefore survives the change of metric and moves to a smaller tile, and the WebP penalty is larger under SSIMULACRA2 than under luma SSIM (-29% vs -8.5% at 224 px): the diagnostic that at equal encoder quality the WebP atlas reaches a lower score than individual files while JPEG does not (for example WebP 112 px q80 scores 72.9 as an atlas against 78.0 individually, JPEG 77.2 against 77.3) confirms that the deficit is the shared-quantizer adaptive-state penalty, and that a perceptual metric weighting chroma and blocking prices it higher.

Table 3. Robustness of the photo crossover under a modern perceptual metric: bytes saved by atlasing (%) at matched SSIMULACRA2 (reference libjxl implementation), for 100 photographic tiles scored on the full grid image at two quality targets. JPEG stays positive at every size; WebP is positive only at 56 px and turns clearly negative by 112 px. The direction matches the luma-SSIM result of Table 1, and the WebP penalty is larger under SSIMULACRA2, which weights the chroma and blocking artifacts of VP8's shared quantizer more heavily. Negative values (red) mean the atlas is larger.

class	JPEG @60	JPEG @70	WebP @60	WebP @70
photos 56px	42.2	33.4	2.8	11.9
photos 112px	17.1	12.5	-21.8	-24.3
photos 224px	4.0	3.0	-28.2	-29.4

The crossover is also a general regime across image populations and codecs, not a property of one corpus. We repeated the matched-quality photo crossover on four independent natural-photo populations (Lorem Picsum and three loremflickr categories: generic, nature, food) and added AVIF alongside JPEG and WebP (Table 4). The sign and magnitude are consistent across populations: at 56 px

JPEG saves 25–27% on every corpus, at 224 px WebP is –5 to –8% on every corpus, and a model trained on any three corpora predicts the held-out fourth to about 4.6 percentage points of error, so a tile-size-and-codec rule, not corpus identity, sets the result. AVIF tracks JPEG rather than WebP, staying positive at every size (+33% at 56 px, +12% at 112 px, +5% at 224 px) and gaining the most at small sizes because its container floor (303 bytes) is the largest fixed cost to amortize. WebP is thus the one codec whose atlas penalty turns negative on large photographs, which sharpens rather than softens the coupling account: the sign of the byte effect is codec-specific, and it is VP8's shared four-segment quantizer, not atlasing in general, that reverses it.

Table 4. Cross-corpus, cross-codec crossover: bytes saved by atlasing (%) at matched SSIM 0.97, median over four independent natural-photo populations (Lorem Picsum and three loremflickr categories: generic, nature, food; 60 tiles each), with the per-corpus range in brackets. The sign and size of the effect are consistent across populations: JPEG and AVIF stay positive at every tile size while WebP crosses zero near 112 px and is negative by 224 px. AVIF, with the largest container floor, gains the most at small sizes. A model trained on any three corpora predicts the held-out fourth's saving to 4.6 percentage points mean absolute error (leave-one-corpus-out over 36 cells), so the crossover is a codec-and-size regime, not a corpus-specific effect. Negative values (red) mean the atlas is larger.

class	JPEG	WebP	AVIF
photos 56px	+26.3 [+25,+27]	+19.2 [+12,+21]	+32.6 [+31,+35]
photos 112px	+8.4 [+8,+9]	+1.6 [+1,+4]	+12.4 [+11,+13]
photos 224px	+2.3 [+2,+3]	-7.0 [-8,-5]	+5.1 [+4,+5]

5.2 Ordering, duplicates, and layout

We re-encode identical tile sets under eight within-atlas orders (source order, three random shuffles, luminance-sorted, mean-color-sorted, k-means cluster-grouped, greedy nearest-neighbor) and two 4-way partitions (random split vs cluster-pure split), for every in-scope codec at $N = 500$. Three calibrated rules emerge. For lossy codecs, ordering is worth at most one percent: the placement of tiles does not recover the shared-model penalty, so the heuristic can ignore order for JPEG and lossy WebP and spend its freedom elsewhere. For lossless flat art, order and partition are real levers: source order (which groups semantically related icons) is already near-optimal, random shuffling costs 2–4%, and cluster-pure chunks beat randomly-split chunks by 3 percentage points (lossless WebP, PNG alike), so same-family tiles belong in the same bundle. For PNG specifically, layout is decisive: packing tiles as a one-tile-wide vertical strip lets the per-scanline filters re-adapt at every tile boundary, turning the PNG flat-art atlas from 5.7% larger than individual files (grid layout) to 8.7% smaller, and lifting the photo atlas from –5.2% (grid, deduplicated 224px photos) to parity. Strip packing is therefore the PNG packing rule (chunked to respect browser image-dimension limits), while block-aligned cells are not worth adopting for any codec: aligning 72-pixel tiles to 80-pixel cells costs 7% in bytes for a quality gain below measurement noise.

The largest partition-level effect belongs to collections containing repeated tiles. Real product grids repeat thumbnails freely (variant images, placeholder art, re-listed items); in a test set where 21.6% of tiles were exact repeats, a similarity-sorted atlas cut PNG bytes by 17–18% and lossless WebP bytes by 15–16% relative to unsorted packing, turning both formats' atlas comparison against individual files from negative to clearly positive (+16.9% and +11.5%). The mechanism is windowed matching: sorting places copies within the codec's LZ window, which dedupes them. Exactly-identical tiles are already handled on the web by content-addressed URLs (the browser fetches one copy and many elements reuse it), so the atlas's unique contribution here is compressing near-duplicates: variant and recolored tiles that share most but not all pixels, which per-URL caching cannot merge. The gain applies precisely to the LZ-class codecs and scales with the duplicate rate, so the heuristic prices it as a measured dup-rate term: deduplicate exactly-repeated tiles at the coordinate-map level (many CSS entries, one atlas region), and sort near-duplicates adjacent so the encoder captures the rest.

5.3 Winning content classes: icons, thumbnails, avatars

The largest bundling wins in the whole study belong to lossless flat art, the content of design-system icon sets, map-marker sprites, and flag pickers. On 200 synthetic flat icons with a small shared palette and alpha (Table 5), a WebP-lossless vertical-strip atlas is the smallest option at every size, 42–68% below individual files and **7.6x smaller than the matched-quality JPEG atlas**. A lossless bundle beats the best lossy one outright, because JPEG must spend heavily to avoid ringing on hard edges while the lossless codecs are both smaller and pixel-exact. A shared-palette PNG (one pooled palette across the strip, verified byte-identical after decode) wins 90–97% over individual paletted PNGs when the pooled palette fits, and remains 4.9x smaller than the JPEG atlas. Strip layout beats grid for WebP-lossless at both sizes, and the duplicate-heavy corpus widens the WebP-lossless strip win from 60% to 68% as the LZ window folds repeats. Two boundaries scope this result: the fixed 12-color corpus is the ideal case for shared palettes, so anti-aliased real icons with hundreds of colors will see a smaller palette win (WebP-lossless, which does not depend on palette size, is the robust default there); and for these alpha-bearing assets JPEG is structurally disqualified, so the operative comparison is PNG versus WebP, both of which the strip atlas improves.

Table 5. Bundle bytes for 200 synthetic flat icons (fixed 12-color palette, alpha), lossless unless noted, on a clean corpus and one with 20% exact + 20% near-duplicate tiles. Lossless bundles are byte-exact (palette conversion verified identical after decode); the JPEG row is the matched-SSIM-0.97 grid atlas. The 90–97% shared-palette saving cited in the text is measured against individual paletted PNGs (per tile), whose byte totals are in the released data. Smaller is better.

bundle	clean 24px	clean 48px	dup 24px	dup 48px
individual WebP-lossless files (baseline)	17,328	23,536	17,050	23,007
WebP-lossless strip	6,966	13,580	5,422	10,606
PNG shared-palette strip	10,900	23,089	6,750	14,855
PNG strip (RGBA)	18,636	38,598	17,802	37,906
WebP-lossless grid	7,780	15,290	7,618	14,910
JPEG atlas (matched SSIM 0.97)	53,266	122,387	51,953	117,377

Two further use cases show where bundling wins on realistic content. Product-catalog thumbnails, which sit on white backgrounds, bundle well under JPEG: a 100-thumbnail atlas saves 13–40% of bytes across 48–112 px, more than the generic full-frame photos of Section 5.1, because the homoge-

neous white field enlarges the fixed-cost share. Under WebP the equal-quality outcome is noisy and not uniformly positive (from +24% at 48 px to -9% at 64 px), because the atlas and individual arms reach nearly identical per-tile quality here, so the byte comparison is dominated by small encoder fluctuations rather than a stable saving. Avatar walls add the duplicate dimension: a comment thread of 200 slots drawn from a heavy-tailed popularity distribution resolves to about 45 unique faces, and an atlas of the uniques (with many coordinate-map entries pointing at repeats) saves 26% under WebP and 50% under JPEG at 32 px versus serving the unique files individually. The lossy codecs do not recover the duplication on their own, an atlas of all 200 slots is 5.0x the size of the unique atlas, so deduplication must be explicit at the coordinate-map level. A methodological note accompanies these: when the atlas and individual arms reach nearly identical per-tile quality, as they do on homogeneous thumbnails, matched-SSIM byte interpolation becomes unstable and equal-quality byte comparison is the appropriate measure, though for such near-ties it too varies with small encoder fluctuations.

6 Discussion

6.1 The coupling spectrum

The results organize on a single axis: what a set of tiles is made to share. At one end, the byte-bundle shares only transport: it concatenates independently-encoded files, so it never costs bytes and never gains cross-image compression. Two standard-container formats sit at the same point, JPEG abbreviated format (one shared table datastream plus per-tile table-less scans) and animated WebP (one independent payload per frame), both measured byte-equivalent to the byte-bundle while offering a playable container or per-tile random access. A shared-table bundle adds table sharing; the pixel atlas goes furthest, sharing the entire coding model, palette, entropy context, quantizer segments, and filters. The measurements then reduce to one principle: bundling *saves* to the extent tiles share *fixed* costs (container overhead, tables, duplicate content), and *loses* to the extent they are forced to share *adaptive* state (a single quantizer allocation, one scanline-filter prediction context, one palette) that individually-encoded files would have specialized. The fixed-cost half of this account is directly testable: regressing each codec's measured saving (63 matched-quality cells) on the overhead-to-content predictor $100 \cdot H \cdot N / \text{bytes}_{\text{individual}}$ gives $R^2 = 0.99$ for JPEG, where the container tables dominate, so amortization alone explains the JPEG savings almost completely (slope 0.59, i.e. the 600-byte figure overstates the recoverable share). For lossy WebP the same predictor captures only the ordering ($R^2 = 0.69$) and badly under-predicts the magnitude, the measured range of -8.5 to +19% dwarfs the predicted 0.5 to 6%, and for the lossless formats the fixed-cost term fails outright ($R^2 < 0.2$, negative for lossless WebP). Those are precisely the codecs where the adaptive-state term dominates: the shared four-segment allocation for WebP, the per-scanline filter for PNG. The coupling account is thus a two-term decomposition, a fixed-cost gain that is quantitatively predictive for JPEG and an adaptive-state penalty that is codec-specific and, at present, characterized rather than modeled; JPEG always wins because its shared cost is large and its adaptive coupling weak, lossless flat art wins because a shared palette is pure fixed-cost saving, and photographic WebP loses under the default encoder because the shared segments are adaptive state, recovering once adaptive deblocking neutralizes the artifact.

A first step toward modeling the second term rather than only naming it: adding a content-heterogeneity feature (the mean pairwise distance of tile thumbnails) with a per-codec penalty coefficient

lifts a pooled cross-codec fit from $R^2 = 0.73$ (fixed-cost term alone) to 0.78, and the learned penalties order the codecs by their adaptive coupling exactly as the mechanism predicts (JPEG 1.2, PNG 5.5, lossless WebP 12.7, lossy WebP 21.0). The modest lift shows a generic heterogeneity proxy captures the direction but not the full magnitude of the penalty.

How far can the saving be predicted from the source images alone, before building any atlas? We tested this directly. Eight content classes were assembled to span extreme image statistics, natural photos, emoji, flags, and avatars alongside synthetic gradients, noise, UI mockups, and objects on white, and six cheap features were computed per class (edge density, DCT high-frequency energy, color-histogram entropy, unique colors, inter-tile heterogeneity, and luminance variance). The content-dependence is real and large: holding tile size and codec fixed, the saving still varies by 16 to 41 percentage points across the eight classes (for example -16% to $+16\%$ at 224 pixels under WebP), so a size-and-codec rule leaves a wide residual. Yet predicting a held-out class's saving from the six features does not improve on that baseline (Table 6), and it fails even on the isolated codec differential that cancels the fixed-cost term (10.1 against a constant's 7.1 percentage points): the adaptive-state penalty is codec-mechanistic and is not captured by these cheap source features. What does recover it is a measurement, not a model: a probe encode of only 10–20 tiles estimates the full-set saving to within about two percentage points, and tracks the content ordering that the baseline misses (within-cell rank correlation 0.86 to 1.00). The probe holds as the group grows, which is where it earns its keep: a fixed 20-tile probe predicts the saving to a mean absolute error of about two points at every group size from 50 to 500 tiles, and orders the tested class-codec cells correctly, so on a 500-tile bundle the decision encodes just four percent of the tiles through the same quality ladder. Its only slip is the WebP near-break-even regime, where the shared-quantizer penalty deepens with group size faster than a small probe registers, the one case a practitioner should confirm on the full set. This is the empirical case for the design of Section 6.2: the right move is not to model the penalty but to measure the two candidate representations, and a small probe suffices, which is why the heuristic reaches the oracle out-of-sample without any content model.

Table 6. Predicting a group's atlas saving before building it, leave-one-content-class-out over eight classes spanning extreme image statistics (JPEG and WebP; 47 of 48 cells, one omitted for no overlapping quality range). Error is mean absolute error (percentage points) against the measured saving; rank is Spearman correlation with it. Six cheap source-image features do not beat a size-and-codec baseline, but a 10–20 tile probe encode predicts the saving to about two points and orders the classes far better.

predictor	MAE (pp)	rank
size + codec + one encode (baseline)	4.95	0.86
+ six source-image features	6.05	0.84
10-tile probe encode	2.31	0.98
20-tile probe encode	1.62	0.98

The spectrum has a hard end. Individual serving costs $\sum_i (H_i + C_i(x_i))$, where H_i is a file's fixed overhead and C_i its compressed payload under state tuned to that image; an atlas costs $H + C(x_1..x_N; \theta)$ with one shared header and one shared adaptive state θ . Bundling wins only when the saved overhead $\sum_i H_i - H$ exceeds the cost of replacing each image's optimal state by the shared θ . For large, unrelated

photographs $H_i/C_i \rightarrow 0$: almost no fixed overhead remains to amortize while any adaptation penalty persists, so a single browser-decodable pixel atlas is not expected to beat independently-optimized files for a sufficiently large, heterogeneous photographic corpus. This is the boundary the measurements trace, JPEG's advantage falling from +30% at 56 px to +3% at 224 px, and it is why the byte-bundle, which shares only transport, is the principled zero-coupling endpoint rather than a fallback. The open direction is to decouple adaptation spatially while keeping single-resource amortization: WebP's lossless format already carries a spatially-varying entropy image (meta-prefix groups), and aligning that grid to atlas tile boundaries, which today requires instrumenting the encoder, is the most promising route to a lossless photographic atlas that wins on bytes.

6.2 A calibrated construction heuristic

The cost above is not minimized by blind search; instead the measured curves calibrate a greedy heuristic, implemented as a command-line tool (`atlas_optimizer`, released with the study). The tool folds exact duplicates into shared coordinates; partitions tiles by update cadence, lossless requirement, and dimensions; and routes each group (Figure 3). For a lossy group the routing is not decided by tile size alone: the tool encodes both a pixel atlas and a byte-bundle and keeps the atlas only if it is strictly smaller *and* passes a per-tile quality gate, so it never adopts an atlas that loses bytes or damages a subset of tiles. The gate is exactly the tail constraint of Section 5.1: the atlas's worst per-tile SSIM must clear an absolute floor (default 0.90) and its 5th percentile must stay within a small tolerance of the byte-bundle's, which carries individual-file quality. On the flat-art set treated as lossy, for instance, a pixel atlas would save 19% of bytes but drives one tile to 0.87 SSIM, so the tool rejects it and emits the byte-bundle instead. Lossless groups take the smaller of a byte-bundle and a WebP-lossless strip, tiny groups stay individual, and every bundle is chunked. It emits the atlas and bundle files, a CSS coordinate map, a loader snippet, and a per-group savings report that records the measured decision and the per-tile SSIM tails. Validated against the study's own asset sets, it converts 521 flat-art tiles into four WebP-lossless strip-atlas chunks (the byte-optimal choice for this lossless class; declaring them lossy-encodable instead routes them to a byte-bundle, since the pixel atlas that would save 19% of bytes fails the quality gate), and 521 photographic thumbnails, 119 of them exact repeats, into four self-describing byte-bundle chunks that collapse 521 requests to four at the deduplicated byte cost (21% below per-reference serving, almost all of it the folded repeats, as on the real directory below). An accounting check confirms the emitted bytes and request count equal what the tool reports.

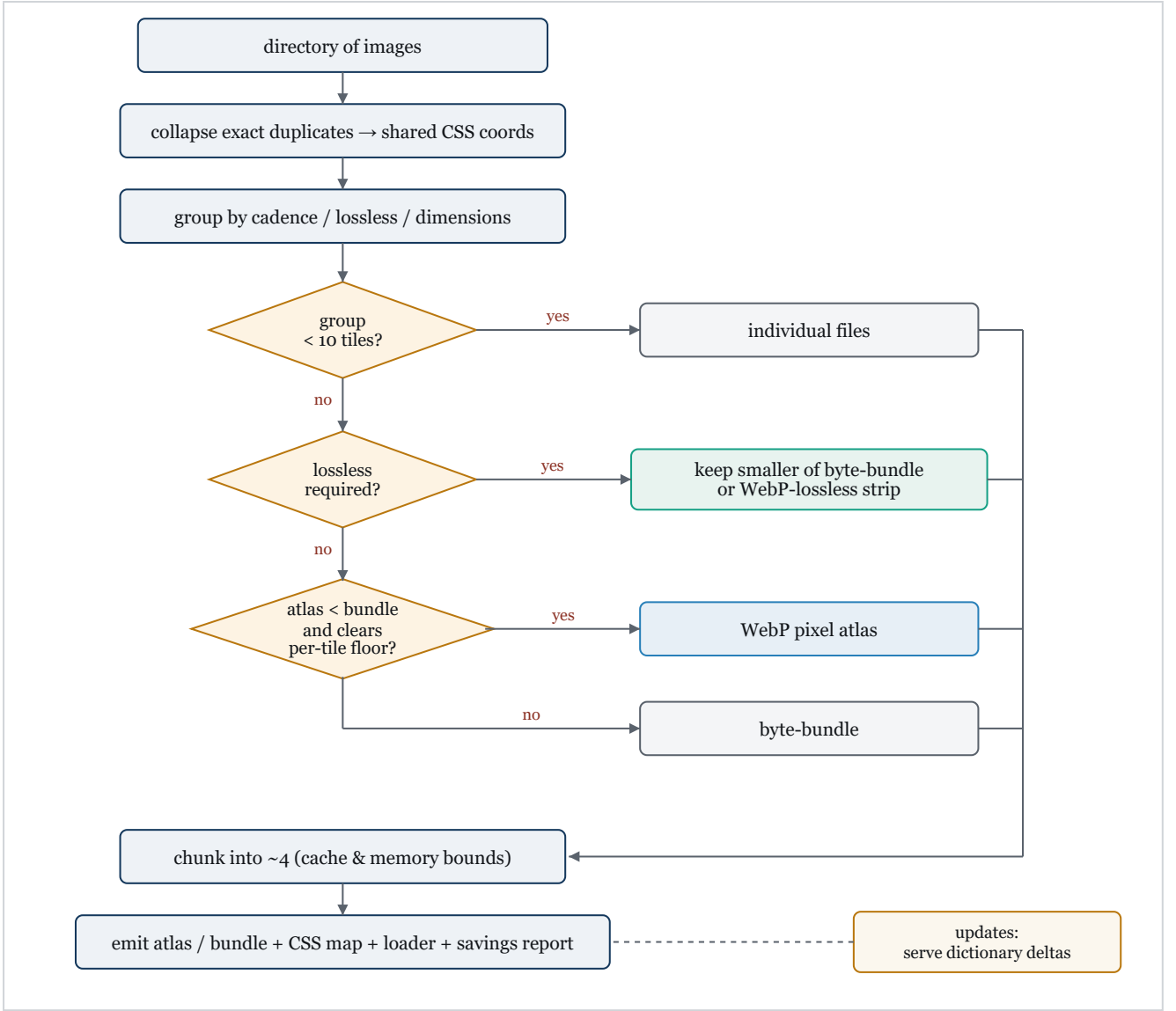


Figure 3. The construction heuristic as implemented in `atlas_optimizer`. Exact duplicates collapse to shared coordinates; tiles are grouped by update cadence, lossless requirement, and dimensions; each group is routed by a short decision cascade to individual files, a pixel atlas, or a byte-bundle. The lossy branch is decided by measurement: a pixel atlas is chosen only if it is smaller than the byte-bundle and clears a per-tile SSIM floor, and lossless groups keep the smaller of a byte-bundle and a WebP-lossless strip. Groups are chunked for bounded cache invalidation and memory, and the tool emits the atlas and bundle files, a CSS coordinate map, a loader, and a savings report; piecewise updates can be served as dictionary deltas. This is the procedure evaluated against the oracle in Table 7.

Because those asset sets also shaped the rules, a fair test requires collections the heuristic never saw. We evaluate it on five independent corpora, Noto emoji, OpenMoji, country flags, Flickr photographs, and Robohash avatars, none of which were used in calibration, against an offline oracle that enumerates the candidate configurations and returns the byte-optimal one (Table 7). Each group carries one measured tiebreak: for a lossless group the heuristic keeps the smaller of a byte-bundle and a WebP-lossless strip, since neither dominates (a strip wins on OpenMoji and Robohash, a byte-bundle on Noto). With that tiebreak the heuristic's automatic choice equals the oracle on all five corpora (0% regret, i.e. zero excess bytes over the oracle's choice), including the two content types, generated avatars and a second emoji vendor, that most differ from the calibration sets. This is the property that matters for a deployable tool: on collections it was not built from, it chooses the byte-optimal configuration. This comparison is on bytes; the per-tile quality floor is an additional safety constraint that overrides a byte-optimal atlas only where it would push a tile below the floor.

The value is in choosing, not in any single layout. Table 7 also scores three fixed rules a developer might hardcode, each still allowed the best admissible codec: always build one pixel atlas, always byte-bundle, or always strip-pack. Every one is beaten on at least one corpus (mean regret 5.8%, 9.4%, and 186%; always-strip alone costs +643% on photographs, where a lossless strip is disastrous), because the byte-optimal layout flips with content: strips win on the sparse OpenMoji and Robohash sets, a byte-bundle wins on dense Noto and on photographs, a pixel atlas wins on the flat-color flags. The calibrated heuristic tracks those flips to zero regret; no fixed rule does.

Table 7. Out-of-sample evaluation of the construction heuristic on five independent corpora that played no role in its calibration (Noto emoji, OpenMoji, country flags, Flickr photos, Robohash avatars; 100 tiles each). An offline oracle enumerates every candidate configuration (individual, grid atlas, strip atlas, byte-bundle across the admissible codecs) at matched quality and reports the byte-optimal one; regret is a policy’s bytes over the oracle’s. The last three columns are simple fixed-layout rules a developer might hardcode, each still allowed the best admissible codec (individual files cost the same bytes as a byte-bundle). The calibrated heuristic matches the oracle on all five corpora; every fixed rule is beaten on at least one, because the winning layout is content-dependent. Smaller regret is better.

corpus	class	tiles	heuristic choice	heuristic	always atlas	always bundle	always strip
noto	flat-art	100	byte-bundle(webpll)	0.0%	+14.3%	0.0%	+5.1%
open-moji	flat-art	100	strip-atlas(webpll)	0.0%	+3.7%	+6.4%	0.0%
flags	flat-limited	100	pixel-atlas(webp)	0.0%	0.0%	+2.3%	+281.2%
flickr	photo	100	byte-bundle(webp)	0.0%	+6.8%	0.0%	+643.0%
robo	avatar	100	strip-atlas(webpll)	0.0%	+4.5%	+38.1%	0.0%
<i>mean regret</i>				0.0%	+5.8%	+9.4%	+185.9%

As an end-to-end check we ran the tool on a heterogeneous directory that mimics a real site’s image folder: 185 files drawn from four live asset sets (country flags, emoji, generated avatars, and photographs) at their native, mixed dimensions, with duplicates included as real pages carry them. The tool folded 29 exact duplicates, partitioned the rest by dimension and lossless requirement, and chose a different representation for each group without any per-group tuning: a byte-bundle for the 224-pixel photographs, WebP-lossless strips for the uniform 64- and 72-pixel avatar and emoji sets, and strips or individual files for the small odd-aspect-ratio flag groups depending on how many share a size. It cut 185 requests to 25. The byte outcome depends on the baseline: 23.5% smaller than serving every file reference separately, but only 3.3% smaller than a deduplicated individual baseline that already serves each unique image once, because most of the first figure is duplicate folding that content-addressed URLs also achieve. The residual 3.3% over dedup is the genuine bundling gain, and it splits exactly as the study predicts: the flat-art strips save up to 11% of bytes while the photographic byte-bundle is byte-neutral (−0.4%) and pays its way purely in collapsed requests. These mixed code paths, which the uniform benchmark corpora never reach, are guarded by the tool’s invariant check: predicted bytes and requests must equal the emitted resources, and the tool passes that check on every asset set in the study.

6.3 Limitations

Quality is measured by luma SSIM at a 0.97 target; the photo crossover is confirmed under the SSIMULACRA2 perceptual metric (Section 5.1, Table 3), and applying it or butteraugli across the full sweep would further sharpen the matched-quality comparison. The shared-palette result uses a synthetic limited-color corpus that is the ideal case; anti-aliased production icons will realize less of it. The network study emulates four profiles on a single-machine testbed rather than the open Internet, and measures time-to-all-tiles-visible rather than a full field-metric suite. Decoded-memory cost is now measured in a live renderer (Section 3.2, Table 9), confirming the pixel atlas is memory-favorable at typical tile counts; profiling under a production framework with texture upload and long-lived navigation remains future work. The HTTP/3 concurrency diagnosis is confirmed from the server's own QUIC transport log (Appendix A.2) and cross-checked against a second QUIC implementation (aioquic), which sustained about twice the concurrency; a fully controlled comparison across server stacks and native Linux remains future work. The matched-quality photo crossover is validated across four independent natural-photo populations (Section 5.1) and the heuristic on five further independent corpora (Section 6.2); a still wider survey of naturally-occurring collections would further generalize the reported thresholds. The network numbers are medians of 11 cold loads per cell with bootstrap confidence intervals (Table 8) but no formal significance testing, and the timing endpoint is time-to-all-tiles-visible; first-tile and first-viewport milestones are reported for the local renderer (Table 9) but LCP, decode CPU, and warm-cache multi-navigation behavior under realistic asset churn are not measured and could change the recommended bundle size. Finally, JPEG XL is left to future work; AVIF is included in the photographic crossover (Section 5.1) and, like JPEG, benefits from atlasing at every tested size.

7 Conclusion

Bundling small web images pays, and the study makes precise when and by how much for the formats that carry most of the web's images. Atlas small lossy tiles: icons and thumbnails gain up to 30% under JPEG, up to 19% under WebP, and up to 33% under AVIF at matched quality, the effect holding across four independent photo populations; and lossless flat art, the icon-set and map-marker case, gains 40–97% with a strip-packed or shared-palette bundle that is smaller than even the best lossy option. Serve larger photographs and lossless assets as a byte-bundle, which collapses requests at near-zero byte cost (a small offset header), or, for photographic WebP, tune the encoder (noise shaping plus adaptive deblocking) to turn the atlas penalty into a gain. Ship about four chunks for bounded cache invalidation and decoded-memory limits, deduplicate repeats explicitly, and delta-encode updates against the cached bundle (Compression Dictionary Transport). On the wire, bundling loads 500 small flat-art tiles 4.5–8.6x faster to full visibility on HTTP/1.1 (1.6–2.8x for photographic thumbnails) and is chiefly a byte optimization on HTTP/2; the comparable HTTP/3 gain reflects a request-concurrency limit specific to the tested server stack (a second QUIC implementation sustains roughly twice the concurrency, Appendix A.2) rather than a protocol-general result. The unifying account, that savings come from sharing fixed costs and losses from sharing adaptive state, is quantitatively predictive for JPEG ($R^2 = 0.99$) and organizes the codec-specific behavior of the rest, and it guides the accompanying construction heuristic, which turns a directory of images into deployable bundles.

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Appendix A

A.1 Encoder parameters

The photographic WebP penalty is not structural. VP8's default encode shares one set of entropy tables and at most four quantizer segments across the whole atlas, and no parameter raises the four-segment ceiling. But two efficiency controls, spatial noise shaping at full strength and adaptive deblocking (autofilter), together move a 120-tile 224-pixel photo atlas from -9.0% to **$+5.0\%$** against individual files at matched SSIM. An equal-quality check isolates the mechanism: with the advanced settings the atlas reaches per-tile SSIM 0.980, above the individual files' 0.979, while encoding 2.4% fewer bytes, because adaptive deblocking removes the block artifacts that the atlas's internal tile-grid boundaries introduce and noise shaping exploits the larger frame's masking context. The knobs improve both arms, but they help the atlas more, so the earlier negative for WebP photographs holds only for the default encoder, not for a tuned one. These controls are reachable through libwebp's configuration (the `cwebp -sns 100 -pass 10 -af` equivalents) though not through every image library's default binding.

A.2 Supporting measurements: network timing

The byte and construction results above are protocol-independent. This section reports how the byte savings translate to load time on a single-machine browser testbed. The deployment guidance draws

on the relative comparisons across conditions, which the randomized, validated protocol holds stable; absolute timings are properties of the testbed.

Table 8. Median time to all 500 tiles visible (ms) per serving condition, protocol, and network profile (n = 11 cold browser loads per cell; 12 run in randomized order, first dropped; WebP payloads). The atl-x column gives the single-atlas speedup over individual files with a 95% bootstrap confidence interval; bun-x is the byte-bundle speedup. Per-cell quartiles are in the released data.

class	network	proto	individual	atlas	atlas x4	byte- bundle	atl-x	bun- x
flat art	localhost	h1	565	126	128	301	4.5x [3.9,5.4]	1.9x
flat art	localhost	h2	448	126	129	319	3.5x [2.3,3.9]	1.4x
flat art	localhost	h3	597	120	135	294	5.0x [4.5,5.7]	2.0x
flat art	100 Mbit / 20 ms	h1	2,074	241	254	466	8.6x [8.5,8.9]	4.4x
flat art	100 Mbit / 20 ms	h2	469	213	216	467	2.2x [2.0,2.3]	1.0x
flat art	100 Mbit / 20 ms	h3	1,579	205	198	431	7.7x [7.0,8.1]	3.7x
flat art	9 Mbit / 60 ms	h1	5,624	790	790	997	7.1x [7.1,7.2]	5.6x
flat art	9 Mbit / 60 ms	h2	790	730	713	1,039	1.1x [1.1,1.1]	0.8x
flat art	9 Mbit / 60 ms	h3	4,020	708	710	1,022	5.7x [5.5,5.8]	3.9x
flat art	9 Mbit / 60 ms / 1% loss	h1	5,827	829	822	1,105	7.0x [6.2,7.5]	5.3x
flat art	9 Mbit / 60 ms / 1% loss	h2	878	832	720	1,098	1.1x [0.7,1.8]	0.8x
flat art	9 Mbit / 60 ms / 1% loss	h3	4,149	867	871	1,189	4.8x [4.7,5.0]	3.5x
pho- tos	localhost	h1	758	424	400	633	1.8x [1.5,1.9]	1.2x
pho- tos	localhost	h2	665	419	422	632	1.6x [1.5,1.6]	1.1x
pho- tos	localhost	h3	835	415	425	662	2.0x [1.7,2.3]	1.3x
pho- tos	100 Mbit / 20 ms	h1	2,019	780	669	1,036	2.6x [2.6,2.6]	1.9x
pho- tos	100 Mbit / 20 ms	h2	692	758	709	1,038	0.9x [0.9,0.9]	0.7x
pho- tos	100 Mbit / 20 ms	h3	2,075	746	763	1,002	2.8x [2.7,2.8]	2.1x

class	network	proto	individual	atlas	atlas x4	byte-bundle	atl-x	bun-x
photos	9 Mbit / 60 ms	h1	6,199	3,824	3,897	4,246	1.6x [1.6,1.6]	1.5x
photos	9 Mbit / 60 ms	h2	3,743	3,771	3,866	4,294	1.0x [1.0,1.0]	0.9x
photos	9 Mbit / 60 ms	h3	6,218	3,902	4,021	4,427	1.6x [1.6,1.6]	1.4x
photos	9 Mbit / 60 ms / 1% loss	h1	6,933	4,612	4,490	7,890	1.5x [1.0,1.7]	0.9x
photos	9 Mbit / 60 ms / 1% loss	h2	5,576	5,993	7,226	6,476	0.9x [0.6,1.3]	0.9x
photos	9 Mbit / 60 ms / 1% loss	h3	8,793	8,404	6,875	7,934	1.0x [0.9,1.3]	1.1x

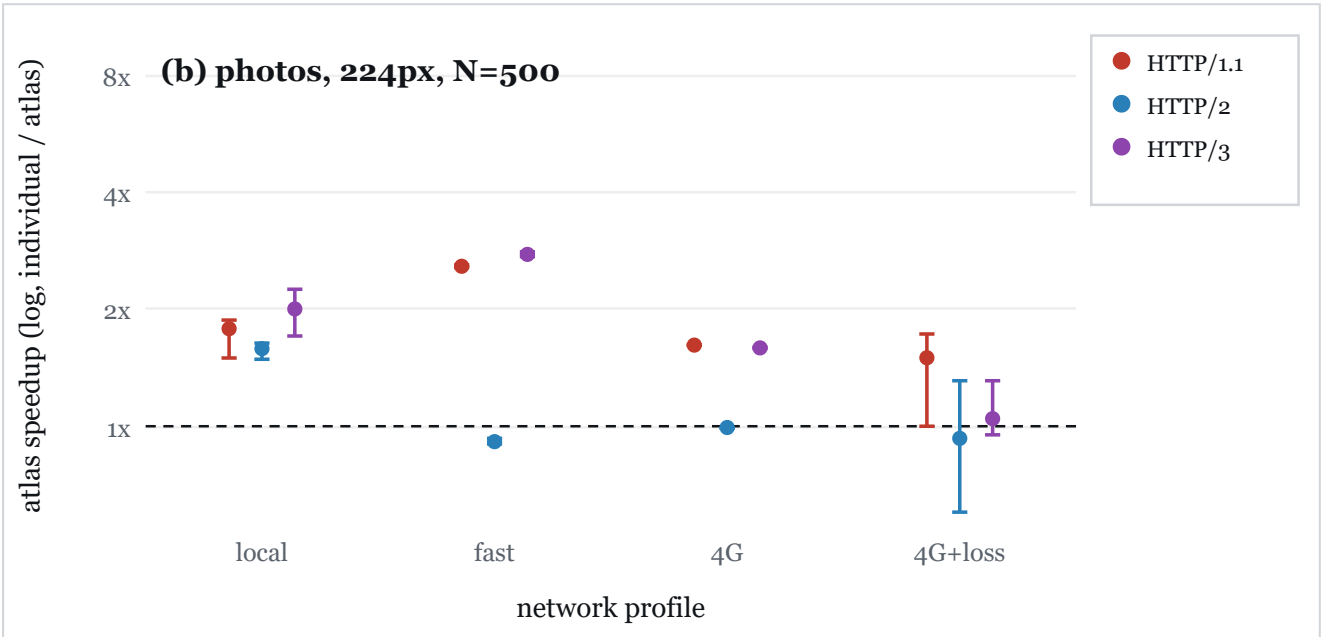
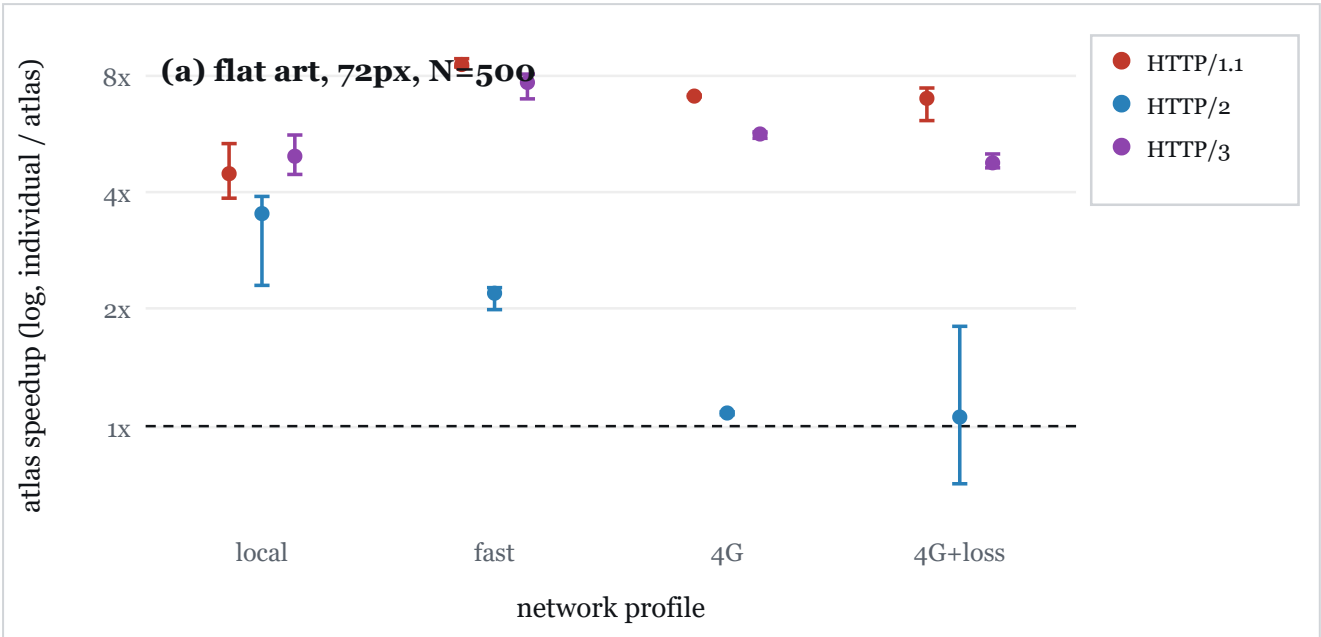


Figure 4. Time-to-all-tiles-visible speedup from atlasng (median individual / median single atlas) at $N = 500$, per protocol and network profile, as point estimates with 95% bootstrap confidence intervals on a shared log axis; the dashed line marks parity (1x). HTTP/2 sits near parity for many small files, while HTTP/1.1 and HTTP/3 retain multi-x speedups; intervals that straddle the parity line (notably the loss-dominated 4G+loss profile) mark cells where atlas and individual are statistically indistinguishable.

Three protocol-level facts organize Table 8 and Figure 4. First, on HTTP/1.1 and HTTP/3, bundling remains a large timing win for flat art: 4.5–8.6x and 4.8–7.7x respectively for 500 tiles across the network profiles, and 1.6–2.8x for photos on the loss-free profiles (the 1% packet-loss photo cells fall to parity and are discussed below). Across cells, added network impairment never speeds a condition up, and the reported speedups carry tight bootstrap intervals (Table 8). Second, HTTP/2 is the strongest protocol for many small files: its multiplexing loads 500 individual images almost as fast as the atlas on bandwidth-limited links (1.0–1.1x at 9 Mbit), so under HTTP/2 the case for bundling small tiles rests chiefly on the byte saving of Table 1 (up to 26% for flat art) rather than on latency. Third, on this testbed HTTP/3's individual-file loads run several times slower than HTTP/2's on the same links, up to 5x for flat art on the constrained profiles and about 3x on the fast zero-loss link, and a concurrency diagnostic locates the cause. The diagnostic reconstructs each request's in-flight interval from the Resource Timing API and finds the peak number of simultaneous image requests: HTTP/2 sustained 13–14 at once while HTTP/3 sustained only 6, roughly half the parallelism, and the slowdown tracks that gap. The server's own QUIC transport log (quic-go qlog) confirms the effect and locates it precisely: across cold HTTP/3 loads of the 500-file page, the server records a peak of only 4–6 request streams open at once (median 5 for photographs, 4 for flat art), agreeing with the browser-side count on the photographic page, while its transport parameters advertise a 100-stream limit (`initial_max_streams_bidi`) that is never approached, so the stream limit is not the binding constraint. To test whether the low concurrency is a property of this particular server, we replayed the load against a second, independent QUIC implementation (aioquic) driven by the same Chromium: it sustained a median of 13 concurrent request streams, roughly double the quic-go/Caddy figure and close to HTTP/2's, on a faster loopback path with smaller tiles that both bias the count downward, so the gap is conservative. The low concurrency is therefore a property of the specific server stack and its defaults interacting with Chromium's QUIC scheduler, not of HTTP/3 itself, which multiplexes independent streams over one connection. A default HTTP/3 stack can therefore under-multiplex a many-small-image page relative to HTTP/2, which makes bundling valuable there, while a different server stack narrows the gap. Under the 1% packet-loss profile every serving condition becomes noise-dominated on this testbed: per-cell coefficients of variation reach 0.4 and the atlas-vs-individual and chunk-vs-atlas differences fall inside overlapping confidence intervals (for example lossy4g photos on HTTP/1.1 give a single atlas at 1.50x and four chunks at 1.54x, indistinguishable). We therefore make no loss-recovery claim for chunking from the timing data; chunking's demonstrated benefit is cache granularity and bounded invalidation (Appendix A.4, Figure 5), not packet-loss resilience. The byte-bundle beats individual serving nearly everywhere on HTTP/1.1 and HTTP/3 (up to 5.6x) at exactly the individual conditions' byte cost, which makes it the bundling method of choice for content whose pixels should not share a codec model (photos, lossless assets); the pixel atlas remains faster where its byte savings compound with the request savings.

A.3 Local rendering cost: decode time and memory

Transport is only half the client cost; decode and memory are the other half, and they are measured locally with no network shaping so they isolate rendering (Table 9). We render N 96-pixel photographic tiles in one Chromium instance three ways and record time to all tiles visible and the renderer process-tree memory over a blank-page baseline, which captures the C++-side decoded-image cache that JavaScript-heap APIs do not expose. At N = 500 the pixel atlas is fastest to full visibility (33 ms, since one decode paints every tile) and uses the least memory (106 MB over baseline), while 500 individual files take 767 ms and 129 MB and the byte-bundle 313 ms and 143 MB, the latter because it fetches its whole object before the first tile and then retains N blob-backed decodes. Two milestones make the progressive difference concrete: the atlas reaches first-viewport-visible in under 25 ms at every N (15 ms at 500 tiles), whereas individual files reach it in 148–429 ms as tiles stream in. For typical thumbnail collections the atlas is therefore memory-favorable rather than a liability, and the width×height×4 caution binds only for very large atlases, which chunking keeps below.

Table 9. Measured browser cost of rendering N 96px photographic tiles three ways, in one Chromium instance with no network shaping: time to all tiles visible (ms) and renderer process-tree memory over a blank-page baseline (MB, capturing the decoded-image cache that JavaScript heap APIs miss). The pixel atlas is fastest to full visibility at every N (one decode paints all tiles) and uses the least memory (one decoded surface rather than N); the byte-bundle must fetch its whole object before the first tile and retains N blob-backed decodes, so it costs the most memory. This refines the analytical width×height×4 bound: for typical thumbnail collections the atlas is memory-favorable, and the decoded-memory caution applies to very large atlases, not to bundling per se.

N	ind ms	ind MB	atlas ms	atlas MB	bundle ms	bundle MB
50	164	89	36	81	44	87
200	379	127	23	92	139	119
500	767	129	33	106	313	143

A.4 Cache invalidation and delta updates

Bundling's one structural liability is cache granularity: editing a single tile invalidates the whole bundle, so a naive re-deploy re-downloads everything. Formally, a partition into bundles $B_1..B_k$ with per-tile update probabilities p_i pays expected bytes $\sum_j [1 - \prod_{i \in B_j} (1 - p_i)] \cdot \text{bytes}(B_j)$ per deploy, which grows with bundle size and with churn. Chunking bounds it at $1/k$ of the collection; grouping tiles by update cadence confines a change to the chunk that carries it. The sharpest remedy is to serve updates as deltas against the cached previous version, the mechanism Compression Dictionary Transport (RFC 9842) now standardizes for the web. We measure the compression-level cost of this remedy as a proxy, computing the delta offline rather than through a live browser-server negotiation: in a 392-photo WebP bundle, a zstd delta against the prior bundle cost 25 KB when 5 tiles changed and 237 KB when 50 changed, tracking the changed-files optimum and beating whole-atlas re-download (2.8 MB) by two orders of magnitude. The structural weakness thus becomes proportional to churn rather than to bundle size; an end-to-end CDT deployment measurement is future work.

A warm-cache simulation over a 200-tile WebP collection makes the full trade-off explicit (Figure 5). Serving immutable individual files is the granularity ideal, a returning client fetches only the changed tiles (12 KB at 1% churn). A single monolithic atlas is the opposite extreme: any change re-downloads the entire 1.47 MB, 126x the individual cost at 1% churn. Chunking closes the gap pro-

portionally, 16 chunks cost 170 KB at 1% churn against one atlas's 1.47 MB, but never reaches per-tile granularity. Dictionary-delta serving does: at 12 KB for 1% churn it matches the individual-file optimum within one percent, and at 20% churn it undercuts it (192 vs 218 KB) because the shared dictionary compresses each new tile against the old ones. The practical consequence is that an update-heavy collection can keep the cold-load benefits of bundling and still get individual-file cache granularity, by serving deltas rather than whole bundles.

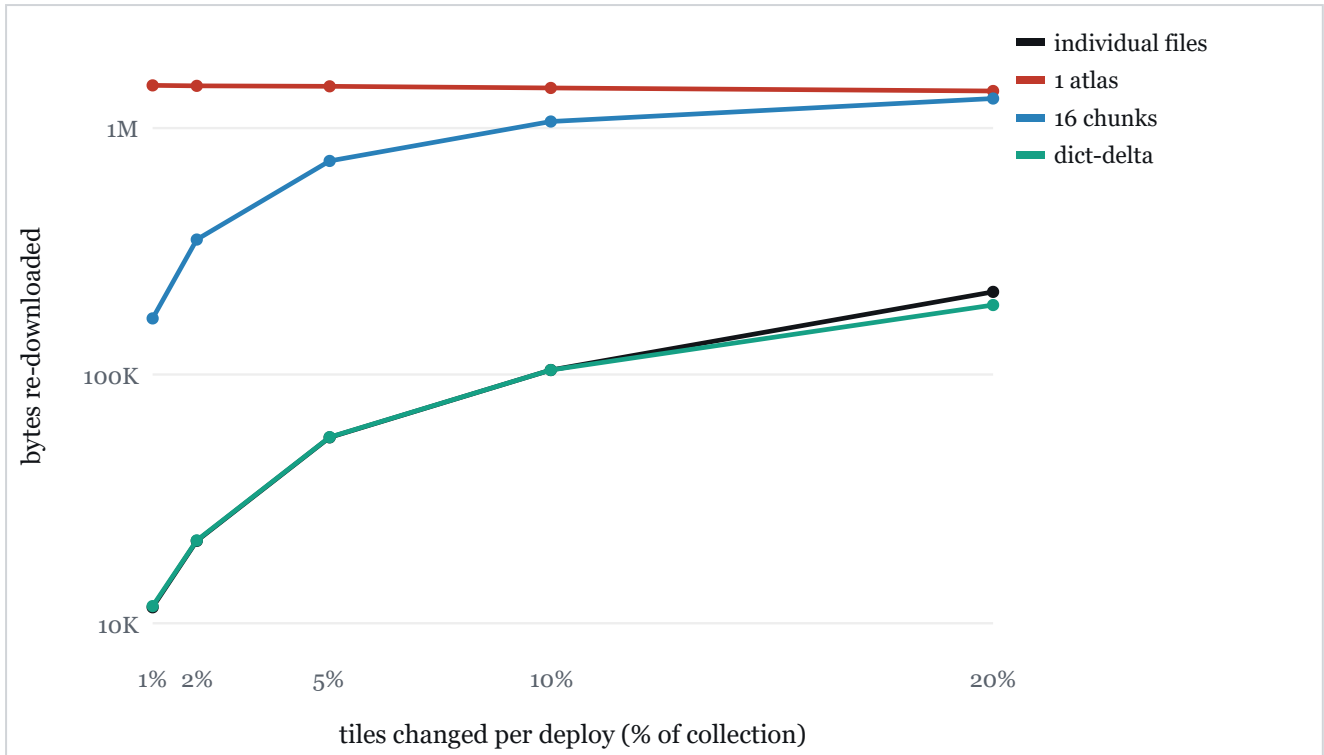


Figure 5. Warm-cache re-download: bytes a returning client fetches after a deploy changes a fraction of a 200-tile WebP collection (log scale; full fresh download 1,474,036 B; mean of 20 random deploys per rate). A single atlas re-fetches almost everything on any change; 16 chunks scale with churn; dictionary-delta serving tracks the individual-file optimum and undercuts it at high churn.

A.5 Choosing a rendering mechanism

Selecting a representation is only half of deployment; the four ways to show a bundled tile (Section 3.1) are not interchangeable in a production page, because they differ in native image semantics, accessibility, loading control, and browser support (Table 10). A CSS pixel atlas shown through `background-position` is universally supported and ideal for decorative icons, but a background image is not an `<img>`: it carries no intrinsic alt text (accessibility must come from surrounding ARIA), does not participate in native lazy loading or `fetchpriority`, and is inappropriate for content images that must be announced. A cropped `<img>` in an `overflow:hidden` wrapper keeps alt text and native loading and works cross-browser; `object-view-box` gives the same with less markup but is Chromium-only and should not be treated as a generally deployable solution. A byte-bundle reconstructs real per-tile `<img>` elements from blob URLs, so it recovers full image semantics at the cost of a small loader script and its content-security-policy implications, which makes it the natural choice for heterogeneous or content-bearing photographs. The selector of Section 6.2 chooses the byte layout; this table chooses the display mechanism, and the two decisions compose.

Table 10. Display mechanisms for a bundled tile and how they differ in production. All four paint after a single atlas or bundle fetch; they diverge on image semantics, accessibility, loading control, and support. Pick the byte layout with the Section 6.2 selector and the mechanism here.

mechanism	native 	alt text	lazy load / priority	JS / CSP	support	best use
CSS background- position	no	via ARIA	manual	CSS only	universal	decorative icons
cropped wrapper	yes	yes	native	CSS only	broad	semantic images
object-view-box	yes	yes	native	CSS only	Chromium only	native crop (emerging)
byte-bundle + blob URLs	yes (after JS)	yes	app-controlled	JS + CSP	broad APIs	heterogeneous photos